Advance Python Assignment

Topic-8: Method Overloading and Overriding

Q1. Explain method overloading in Python.

Ans.
Method overloading refers to defining a method with the same name but handling different types or numbers of arguments. Python does not support traditional method overloading, but we can achieve similar behavior using default arguments or variable-length arguments. This allows a single method to perform different tasks based on the input. It makes the program flexible and reduces the need for multiple method names.

Example:
```
class Calculator:
    def add(self, a, b=0):
        return a + b

c = Calculator()
print(c.add(5))
print(c.add(5, 10))
```

Q2. Explain method overriding in Python.

Ans.
Method overriding occurs when a child class provides its own implementation of a method that is already defined in the parent class. It is used to change the behavior of the inherited method. When the method is called using the child class object, the overridden version is executed. This concept is useful when we want to provide specific functionality in the child class.

Example:
```
class Shape:
    def draw(self):
        print("Drawing shape")

class Circle(Shape):
```

```
    def draw(self):
        print("Drawing circle")

obj = Circle()
obj.draw()
```